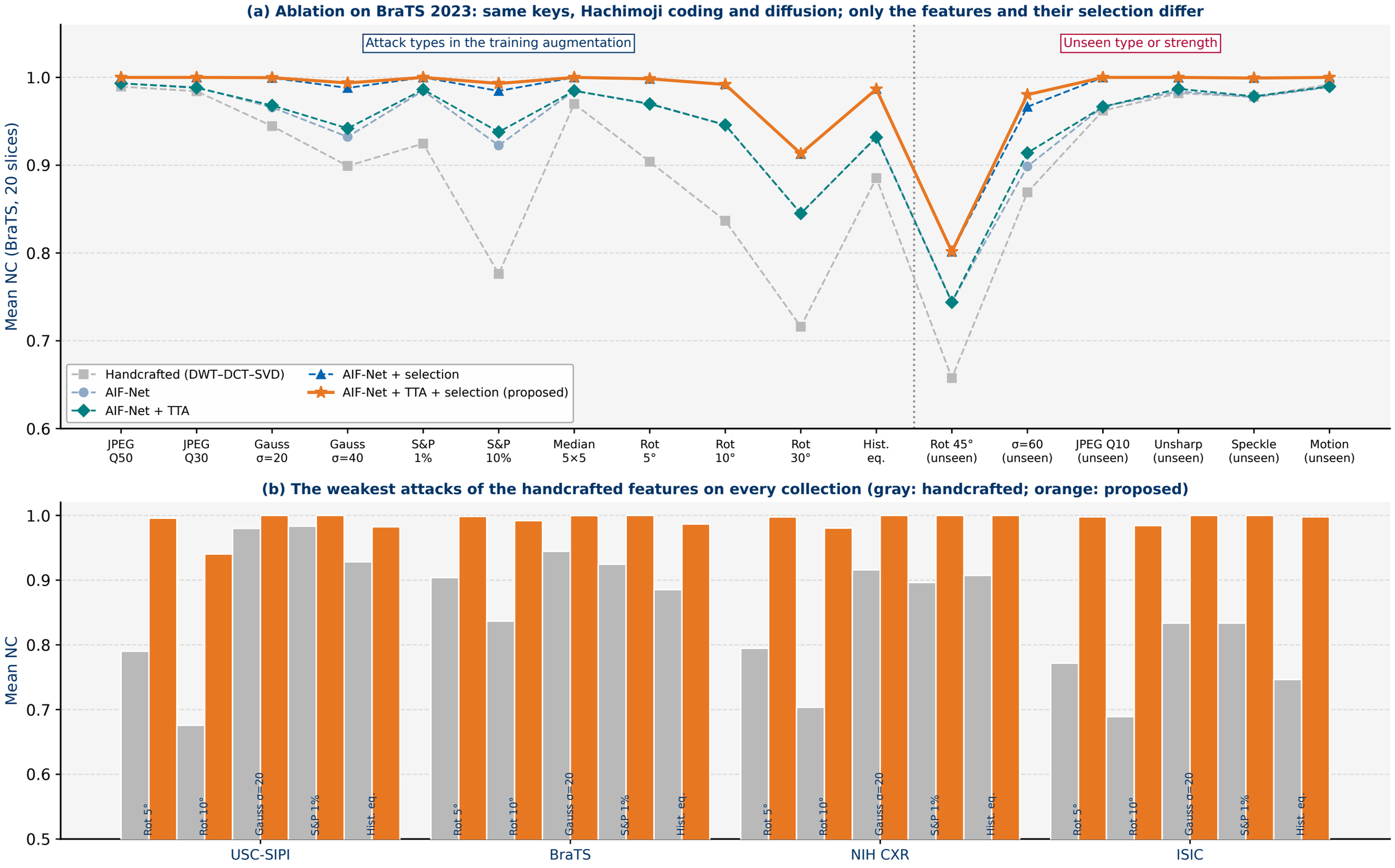
(a) Ablation on BraTS 2023: same keys, Hachimoji coding and diffusion; only the features and their selection differ
Attack types in the training augmentation
Unseen type or strength
Mean NC (BraTS, 20 slices)
1.0
0.9
0.8
0.7
0.6
Handcrafted (DWT–DCT–SVD)
AIF-Net
AIF-Net + TTA
AIF-Net + selection
AIF-Net + TTA + selection (proposed)
JPEG Q50
JPEG Q30
Gauss σ=20
Gauss σ=40
S&P 1%
S&P 10%
Median 5×5
Rot 5°
Rot 10°
Rot 30°
Hist. eq.
Rot 45° (unseen)
σ=60 (unseen)
JPEG Q10 (unseen)
Unsharp (unseen)
Speckle (unseen)
Motion (unseen)
(b) The weakest attacks of the handcrafted features on every collection (gray: handcrafted; orange: proposed)
Mean NC
1.0
0.9
0.8
0.7
0.6
0.5
Rot 5°
Rot 10°
Gauss σ=20
S&P 1%
Hist. eq.
USC-SIPI
BraTS
NIH CXR
ISIC